Venkata Lakshmi Parimala Pasupuleti

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PROFESSIONAL SUMMARY

AI engineer and solutions architect with **5+ years of hands-on solutions architecture experience** - designing and delivering production agentic systems, RAG pipelines, and enterprise AI architectures for compliance, operations, and finance stakeholders. Architected **Leasel** (8-agent LangGraph platform serving three enterprise surfaces) and **Magizhs** (agentic mission-planning system deployed across three commercial verticals) - both self-directed from requirements definition through production delivery. Deep **LangChain/LangGraph/LangSmith** expertise: multi-agent orchestration, cognitive architectures, RAG, HITL, confidence scoring, and agentic process automation. Delivered **live technical workshops to 600+ stakeholders** at George Mason University. M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026).

CORE COMPETENCIES

Agentic AI Architecture | LangChain / LangGraph / LangSmith | RAG Pipelines | Enterprise Solution Design | Conversational AI, Contact Center & HITL | Technical Workshop Delivery | Agentic Process Automation | Multi-Stakeholder Technical Advisory

WORK EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI Solutions Architect - Agentic Systems (Intern)

- **Architected and delivered an end-to-end agentic AI solution** for drone mission planning - defined requirements, designed the cognitive architecture, and deployed to production across marketing, real estate, and commercial survey verticals without a spec doc or external direction.
- Solution architecture: LangGraph stateful graph (parse → plan → retrieve → validate → conditional routing), RAG layer over FAA Part 107 regulatory corpus, confidence-scored outputs, validation guardrails enforcing altitude limits and no-fly-zone checks.
- Delivered solution as a working POC across three distinct customer verticals - adapted architecture for different use case constraints in each deployment; improved agent accuracy through iterative prompt engineering.

George Mason University - College of Engineering

Aug 2025 - May 2026

Graduate Teaching Assistant - Technical Workshops

- **Designed and delivered live technical workshops to 600+ students** - Python programming, AI fundamentals, and database design; translated complex technical architectures into hands-on exercises for non-technical audiences.
- One-on-one technical advisory sessions; rubric-aligned grading and structured feedback - the same communication and stakeholder translation skills enterprise SE work demands.

George Mason University - Research

Aug 2024 - Aug 2025

Research Associate - AI & Data Analytics

- Built multi-source prediction models (Python, R) across 190+ countries - R^2=0.977; designed Tableau/ggplot2 dashboards improving research comprehension 40%; applied Isolation Forest anomaly detection across 3 research pipelines.

Entrata - PropTech / Real Estate SaaS (100K+ MAU)

Sep 2018 - Aug 2024

Senior Software Engineer

- **5+ years as embedded solutions architect** to compliance, finance, and operations stakeholders - translated domain requirements into solution architectures across 120+ Agile sprints; defined integration patterns, data models, and system boundaries for 20+ production modules; zero compliance violations over 6 years.
- **Architected and delivered enterprise solutions** (PHP, Python, React, PostgreSQL, RabbitMQ, AWS): income eligibility engines, AMI-based pricing logic, HUD compliance reporting, HAP allocation workflows - 99.5% uptime sustained across 100K+ MAU product.
- **Designed AI solution architecture** for first production AI pipeline on 1.6M financial records - scoped requirements, designed validation gates and compliance guardrails, delivered to production with zero incidents (Python, PyTorch, OpenAI API).
- Led microservices redesign (Docker) - 50% scalability improvement, 30% deployment time reduction; AWS S3 migration with 100% PHI compliance; mentored 5+ engineers across 300+ peer code reviews.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer / Trainee

- Built full-stack property management modules (PHP, Python, PostgreSQL); ETL pipelines for 1.6M HUD/Tax Credit records; cloud migration to AWS S3; redesigned document storage database architecture.

SOLUTION ARCHITECTURE - REFERENCE IMPLEMENTATIONS

Leasel: Enterprise Agentic AI Platform - Document Intelligence & Support Automation 2026

Designed and built a **production-grade agentic solution architecture** for affordable housing compliance operations: 8-agent LangGraph cognitive architecture (Triage → Retrieval → Configuration → Historical → Reasoning → Validation → HITL Escalation) with real conditional routing, confidence scoring, and hard safety boundaries enforced in code. **Three enterprise surfaces**: Document Intelligence (RAG retrieval, 91% Precision@1, 100% citation correctness), Support Investigation (automated root-cause analysis, 93.6% precision), and Tenant Self-Service (agentic process automation with human escalation). Reproducible eval harness: 225-query benchmark, LLM-as-judge scoring, regression gating.

Python · LangGraph · LangSmith · FastAPI · RAG · HITL · Agentic Process Automation · LLM-as-Judge eval

AI on FHIR: NLP Solution Architecture - Clinical Data Retrieval 2024

Architected and delivered an NLP platform translating natural language queries (EN/ES) to FHIR APIs using LLMs, BERT, and embeddings - reduced manual clinical retrieval time ~60% and enforced HIPAA-compliant RBAC across structured medical records.

Python · Flask · LLMs · FHIR · React · spaCy · BERT · Embeddings · Docker

EDUCATION

M.S. Data Analytics Engineering · George Mason University, Virginia — May 2026

GPA: 3.87

B.Tech Computer Science & Engineering · Jawaharlal Nehru Technological University — 2017

SKILLS

Languages: Python, JavaScript, TypeScript, SQL, PL/SQL, PHP, R, Bash

AI & Agentic Architecture: LangChain, LangGraph, LangSmith, Agentic AI, Cognitive Architectures, Multi-Agent Orchestration, Agentic Process Automation, Conversational AI, RAG Pipelines, Tool Calling, A2A, MCP, HITL, Guardrails, LLM Evaluation, LLM-as-Judge, Confidence Scoring, Prompt Engineering, Hallucination Detection, OpenAI API, Anthropic API, FastAPI, ChromaDB, spaCy, BERT

Cloud & Integration: AWS (S3, EC2, RDS, Glue), Azure, Docker, Microservices, API Integration, CI/CD (GitHub Actions), PostgreSQL, MySQL, Oracle, RabbitMQ, Git, Linux, Agile/Scrum

Data & ML: PyTorch, Scikit-learn, XGBoost, Isolation Forest, Feature Engineering, Pandas, NumPy, Tableau, Power BI, ggplot2